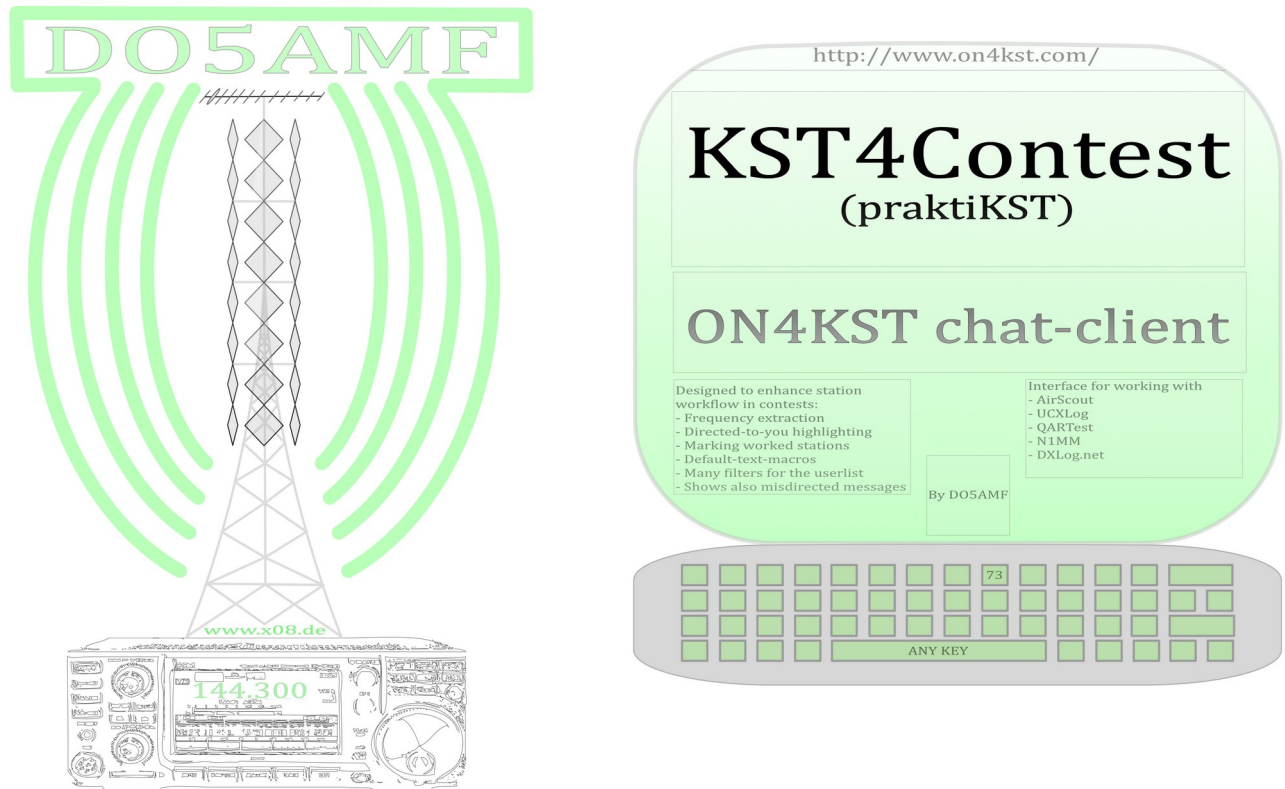

Chat mining: awakening hunting instincts with KST4Contest

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1) Foreword

Historical note: this document records the development and knowledge available at GHz-Tagung Dorsten 2026. In 2024, Ralf, DG6EA, noticed my work on the ON4KST chat and the power supply at our contest station DM5M, and together we settled on the subject of an ON4KST chat client. I was invited to speak about it at GHz-Tagung Dorsten in 2025. The subject was received surprisingly well and prompted intense discussion. Some of the ideas and the roadmap I presented for the program caused a murmur in the room. That made further development of certain functions seem welcome from the outset - while ruling out others.

A slightly indignant murmur entered the room when I mentioned that stations in the chat could be contacted automatically if we had not worked them yet. After that reaction, I decided to remove the feature from the list. I can reassure everyone: it has stayed there.

This is the paper accompanying my presentation in Dorsten in 2026. The paper from the previous year is available in the 2025 GHz-Tagung proceedings. It explains design decisions and background on which this new presentation builds - and has to build, because a complete overview would exceed the scope of this paper.

I hope the kindly disposed reader has not forgotten everything covered in the first part of my presentation. If so, it can always be read again; Ralf probably still has copies from last year.

This year's title cost me some sleep because, at first, I did not know exactly where the subject would lead or what results could be expected from the actual implementation. The implementation was based mainly on theoretical considerations arising from the chat analysis. Turning them into concrete features is sometimes complicated, and I continue to work on it. By the time of the presentation, the scope and appearance of the application will already differ from what is shown here, because this paper had to be finished about four weeks before the conference.

I hope you enjoy reading it and attending what will be my second GHz-Tagung in Dorsten in 2026.

73 de DO5AMF, Marc

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2) Chat mining - what is that supposed to be?

Data mining sounds unwieldy, but everyone has heard of it somewhere. By definition, it means the automated analysis of large data sets to find patterns, correlations and useful information. In IT, the aim is to optimise processes. As we know from last year, that is the main purpose of the KST4Contest chat client.

I explained last year that metadata and contextual information from the ON4KST chat can be valuable sources of information, and the previous version of KST4Contest already used some of them. Certain information can be used directly to provide value while the chat client is running. The most immediate example is the text itself, from which we can extract a frequency.

Other information provides value indirectly because it lets us derive details that are not immediately obvious to the chat user. One example is the operating interval of a particular contest station.

Finally, some information is interesting simply because nobody else has looked at it. Perhaps that, too, offers some value. If not, it is at least entertaining to read. When larger amounts of data accumulate, being a computer scientist compels me to investigate. In the KST chat we are talking about 130,000 lines of text and more. Surely there must be something else to extract from it.

A sound data source is needed. Fortunately, it is openly available. Chat logs remain accessible on the main www.ON4KST.org website for about a year. Under “Menu” → “chat review”, the complete chat history for a selected date can be retrieved as text. The text can also be filtered, for example by callsign or text pattern. I did not do that because I needed the complete data set for the analyses.

I analysed the chat logs from IARU contests over the preceding two years, including the July 2025 contest. I use that contest here to demonstrate the information that can be derived. The numerical relationships led to very similar conclusions for all contests. Absolute figures sometimes differ considerably, but the trends are alike.

I deliberately chose the July 2025 contest for the concrete figures because I could combine data from two chat categories - 144/432 MHz and Microwave - and look for correlations across those categories.

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3) A matter of definitions

Before drawing conclusions from data, one has to decide what to look for. As my algorithms and data structures professor, Dr Wolfgang Schramm at Mannheim University of Applied Sciences, used to say: “It starts with a blank sheet of paper.” I am used to that by now, so I defined a few parameters for the analysis. They were...

a) Where they were

Which chat was a callsign in? Did it change chat channels? How often? Where did it write?

b) How long they stayed

How long was the callsign present in total? For how long were stations reachable in the chat at all? Can operating periods be inferred from those intervals?

c) Success and failure in the chat

There are unambiguous key phrases that can show that a sked was successful even without the contestants' QSO logs. Examples include “tnx, 73”, “tnx qso” and “tnx for xxx points”. There are equally clear phrases showing that a sked was unsuccessful, such as “tnx try”, “no copy” and “no ping, sry”.

I first try to measure the success or failure of a sked attempt without reliable explicit information about a station's intention or the beginning and end of its sked. That is usually almost impossible, because nobody wants to do extra documentation purely for statistics. I therefore started with text analysis:

- **Evaluate the sked phrase and recognise the intention, for example “try sked 144.210?”.**
- **Open a sked time window, which I will later call an “init” or “window”.**
- **Evaluate a phrase that indicates the sked result, then close the sked time window or terminate it if no closing message can be found.**

In some cases this works. In most cases it does not. One reason is that success can be derived from chat phrases only very crudely: most contacts are naturally confirmed and completed over the air. It gets worse. There are considerably more requests than replies. A large number of stations contact others periodically and either transmit immediately - which our experience shows is usually what happens - or, much more often, receive no reply. The recipient may have no time or no interest.

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Perhaps they already use KST4Contest and do not need periodic invitations to other stations' QRGs, or perhaps they have tried five times without success and now ignore the requests.

Later in the text analysis I also try to determine whether mass mailings - identical messages sent to a list of stations - make sense. Put another way: are “spammers” more successful than stations that use their messages more sparingly and deliberately? Quality over quantity? When these lines were written, the result was not yet known, and I hoped I would obtain one.

Intuitively, inviting unworked stations to one's own QRG produces good results. At DM5M this works particularly often with stations in 9A and G. They do not reply in the chat; they come to the stated QRG and try. I have seen the reverse while operating as DO5AMF/p with three elements and 10 W on the Baltic coast, with stations such as OE1W. Ten sked requests stating their QRG arrived within an hour. I politely declined the first ones. They kept coming. I was QRV for about six hours and received more unnecessary messages from OE1W than from anyone else.

It seems particularly sensible to contact other stations when the antennas really are pointing towards them and the path can work. Excessive messages that ignore path conditions or even antenna direction will quickly make a station ignore the sender after several failed attempts - even if Aircraft Scatter conditions later improve and the path becomes possible.

d) Chat efficiency

I touched on this subject in my previous presentation. One can debate the sense and nonsense of using chat on VHF; it is often condemned outright. Perhaps I am adding fuel to the fire, or perhaps doing the opposite, by trying to make the effects of chat use visible. Measuring “positive signals” such as replies containing “tnx qso” is quickly exhausted and says little because such messages are too rare. The next step is therefore to look for correlations between the scoreboard and the chat, assess the efficiency of using the chat and derive possible improvements.

Enough dry introductory material. I will now present some interesting data produced by these analyses, along with occasional conclusions for our own operating practice and direct consequences for KST4Contest, before turning to the new functions.

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4) Some interesting figures from one example

I explicitly use the IARU VHF contest of July 2025 here. The results and statistics below are very similar for the other IARU VHF contests, which allows broader statements to be considered.

a) Number of messages

The July 2025 contest took place on VHF and extended upwards with almost no practical frequency limit. There were 117,399 messages in total: 111,206 directed messages and 6,139 undirected messages without an explicit recipient, which I call “CQ” or “broadcast”. The Microwave chat leaned even more strongly towards addressed messages: 97% were addressed, compared with 93% in the VHF chat.

This differs markedly from the September 2025 contest, in which there were far more broadcasts. A large part of them resulted from invalid syntax in messages intended as private messages; see the 2025 paper. They can be recognised by phrases such as “CQ <Callsign>” without the slash, or “(Callsign) <message>”.

Many chat clients do not highlight those messages, so the intended recipient probably does not read them. In September, such messages accounted for about 8% of the total, closely matching the pattern from September 2024. Interestingly, in NAC and UKAC contests the proportion of “broadcasts that should not have been broadcasts” is about twice as high.

I had already addressed the problem in the previous year and solved it in KST4Contest by processing such messages as private messages.

b) Number of participants

There were 1,172 unique callsigns in the chat during the contest's 24 hours across VHF and Microwave. Category 2, 144/432 MHz, contained 964 unique callsigns; category 3, Microwave, contained 411. Category 2 but not category 3 contained 765 callsigns as senders and recipients. Category 3 but not category 2 contained 210. A total of 144 changed between categories, although this can be measured only for senders.

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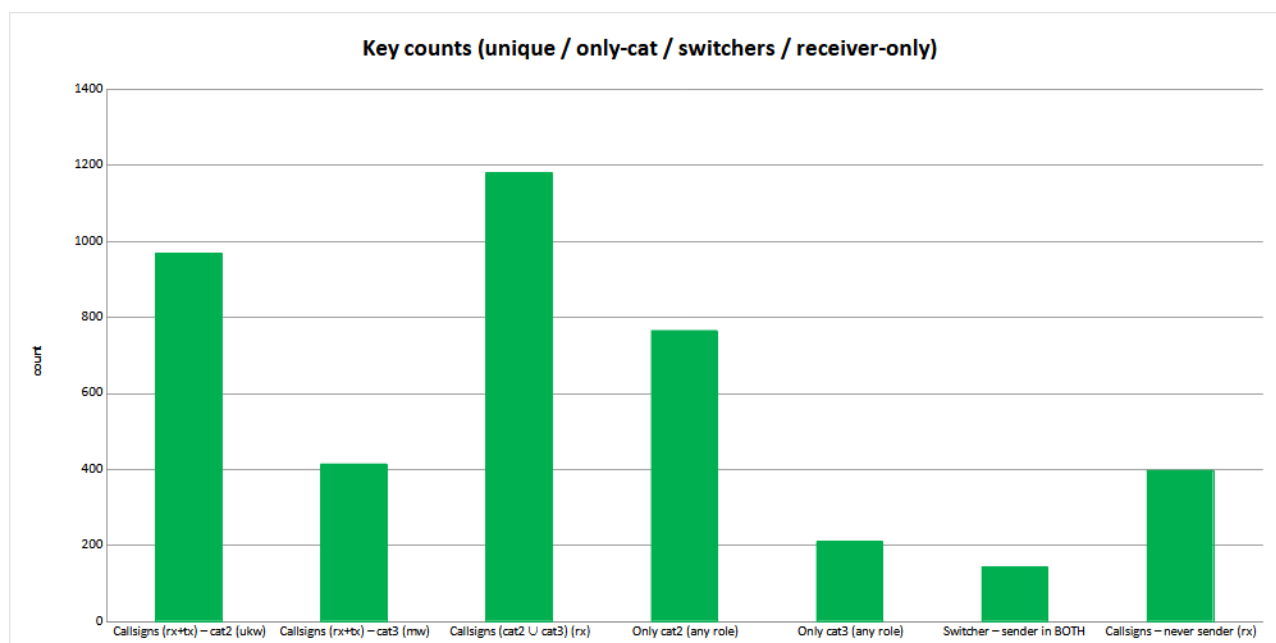


Figure 4.1: Unique callsigns in the individual categories, by receive/transmit activity and category4.1

Figure 4.1: Unique callsigns in the individual categories, by receive/transmit activity and category. This gives us an indication of whether observing both categories is worthwhile. As soon as an operator combines VHF with 23 cm or higher bands - or the other way round - the answer is clearly yes. Stations active on 23 cm and above do not always enter or monitor the 144/432 MHz chat. Formally, that makes sense. It also means they have no opportunity to attract sked partners such as DM5M, where 23 cm is treated as an “already fitted” band because the IC-9700 includes it and we do not want to leave it unused.

Fortunately, software can help with this. In March 2025 I added a second category to KST4Contest using Single Sign-on. KST4Contest is not the only client able to do this; the VUSC4WIN logger can as well. That logger also supports the much-loved mass mailings.

c) Senders and recipients

A total of 397 callsigns never transmitted and only listened. In category 2, 355 callsigns only received; in category 3, the corresponding figure was 94.

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The fact that 355 plus 94 does not equal 397 is not a contradiction. Some callsigns in those subsets transmitted in one category but only received in the other. The 397 callsigns are those that appeared solely as recipients everywhere.

Conclusion: many stations reply in only one of the two categories. And because these stations are not in the chat 24 hours a day, 365 days a year - only about 40 are - they must have some interest in receiving messages even when they do not reply. It apparently remains worthwhile to contact callsigns that do not answer. In some contests, the rules do not allow certain stations to reply, as Tim, DK5OH, explained to me some time ago.

d) Activity peaks over time

The peak in message volume occurs on Sunday from 07:00 UTC. This is similar in all IARU contests I analysed. It suggests the best period for a six-hour entry that relies heavily on skeds. Operators such as SM6VTZ and SK7EYW are often QRV during this period in the six-hour class. Besides Facebook, a useful source of participation information is <https://qrvmmap.vhfdx.ro/>. Figure 4.2: Messages per hour in the chat channels

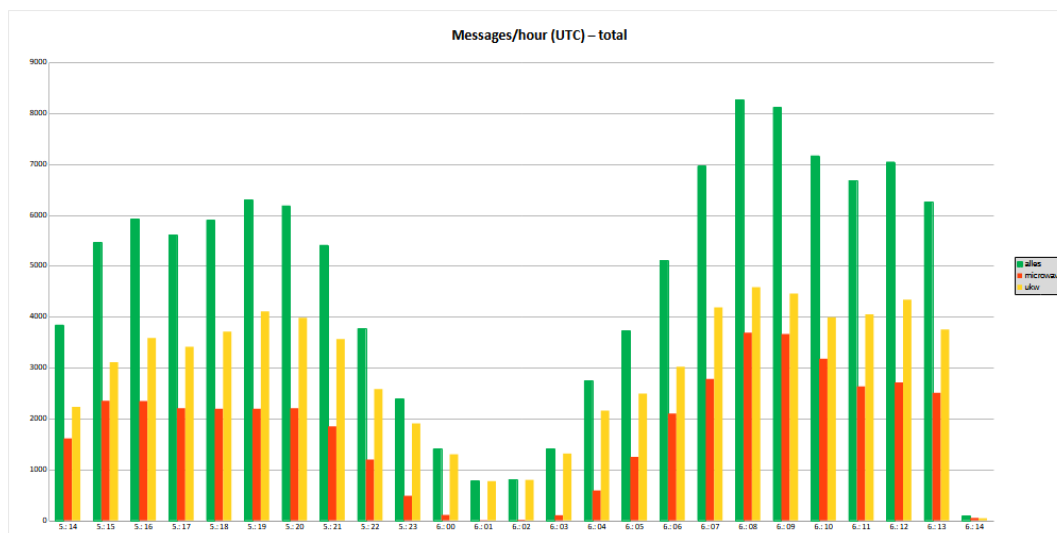


Figure 4.2: Messages per hour in the chat channels4.2

This finding is slightly deceptive, so I will state the distinction explicitly. Contest activity is clearly concentrated near the beginning of the contest, as the typical QSO rates in the first hour already show. Chat activity, however, peaks on Sunday at around 07:00 to 08:00 UTC.

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One might suspect that on Saturday the bands are busy enough for skeds to be less necessary. Contacts with the strongest signals are made first. Difficult QSOs that need preparation and cannot be completed simply by calling CQ or scanning the band are scheduled for the night or Sunday.

Weak stations and stations outside central Europe depend on skeds, so they probably concentrate their activity on Sunday. From experience: when I operated a VHF contest from SM in July 2020, almost nobody happened to turn their antenna north. Friends in SM and LZ have reported similar experiences.

For precisely that reason, these operators repeatedly asked for a QSO sniffer that would allow messages to nearby stations to be followed deliberately. They could then use an opportunity when a potential counterpart turns its antennas towards them for a sked. I implemented the QSO Sniffer in December 2025. It allows messages addressed to selected callsigns to be monitored.

I had implemented “Sked directed to me” much earlier. Its purpose is similar, but it uses more heuristics and demands less operator attention. The feedback has been consistently positive.

e) Big guns in the chat

On the bands, it takes some hardware to make oneself heard. A familiar approach is to increase not only the number of antennas but also the number of directions they cover, and of course to add power. In combination, the station attracts attention and becomes a “big gun”. One can also attract attention in the chat. The required investment is more modest.

The ON4KST chat itself provides no built-in means of measuring a participant's visibility, so I examined the subject more closely.

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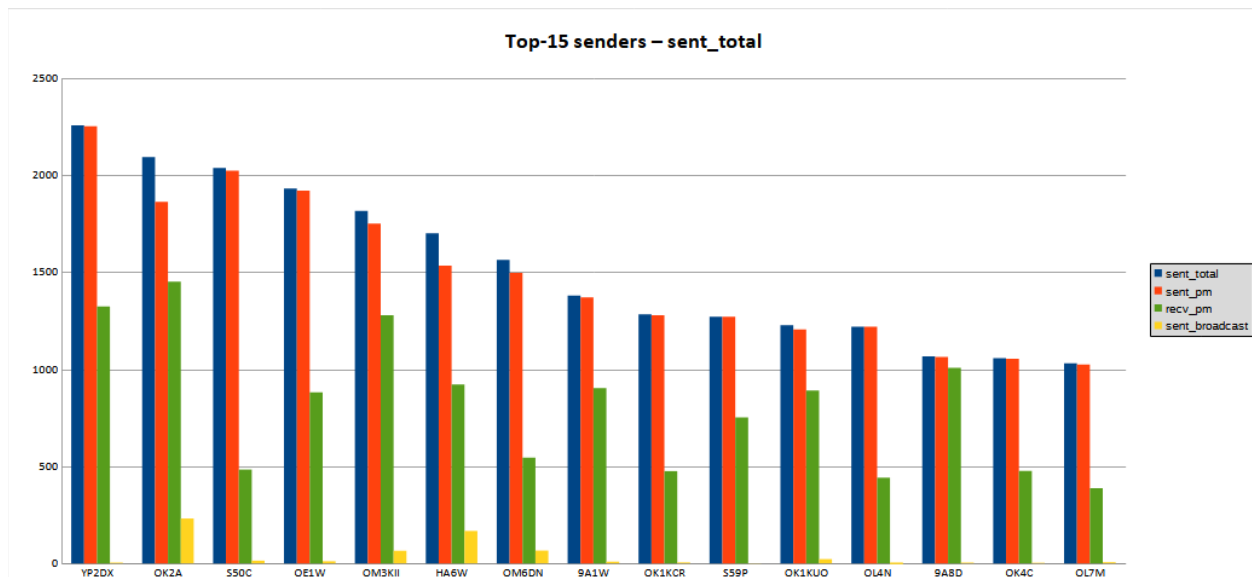


Figure 4.3: Top 15 message senders and replies to their private messages^{4.3}

Further analysis showed that S50C sent the most messages per hour on average. The operator at the keyboard certainly had stamina. OE1W followed directly, but with a good 30% fewer messages on average. The diagram does not show the messages-per-hour pattern; it shows only the absolute figures for the top 15 senders. It also shows the number of private messages received in relation to those sent, which gives an impression of whether skeds resulted. Figure 4.3: Top 15 message senders and replies to their private messages

f) Success rates

1) Relationship between contest score and chat use

For some time I have wanted to understand more precisely how using the chat might affect contest results. My first attempt was to ask: “How many stations in the top 20 of each entry class did not use the chat?” There is an answer, but it is not enough.

A meaningful view must also consider the number of entrants in each class. If there were only ten, all of them would be in the top 20. Rather than analysing fixed places 1 to 20 for chat use, the number of top placements must therefore be normalised against the number of chat users.

Omitting this normalisation would make the results misleading and would not meet the standard of a neutral analysis. I learned that while studying computer science 15 years ago and then forgot it again. Now the terminology has returned. Greetings to Professor Dr Frank Nürnberg: it was not all

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in vain. I deliberately place three diagrams one after another before offering a conclusion.

Figure 4.4:
Submitted logs
without chat use,
by entry class

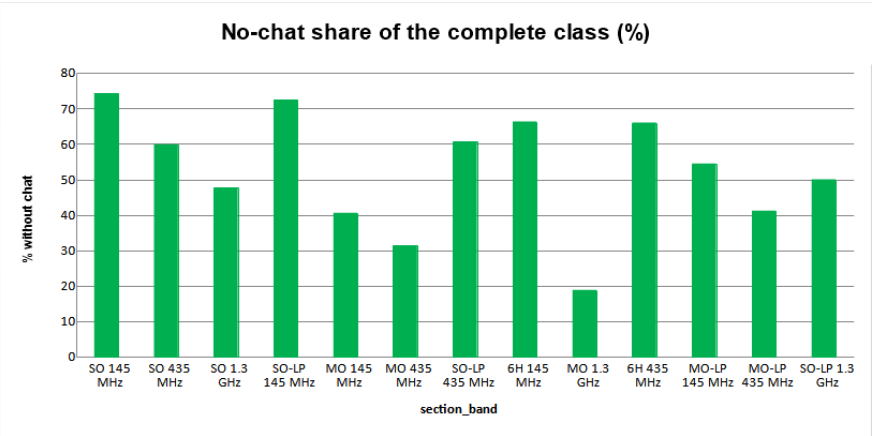


Figure 4.4: Submitted logs without chat use, by entry class4.4

Figure 4.5: Total
submitted logs by
entry class. Figure
4.6: Logs without
chat use placed in
the top 20, by
entry class

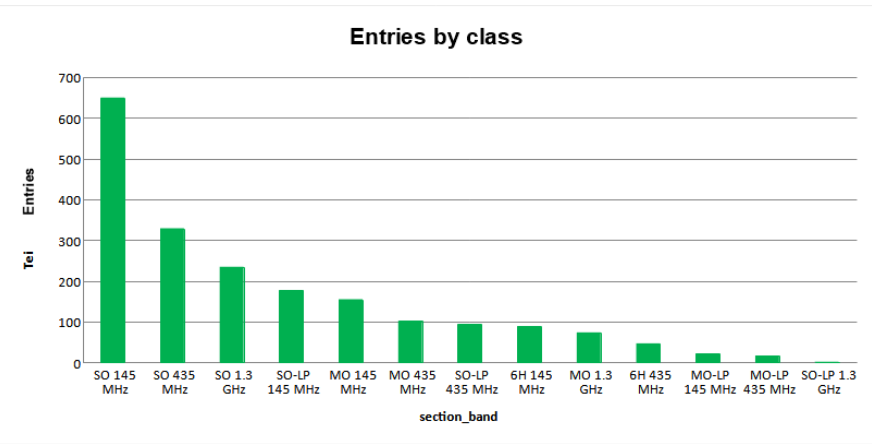


Figure 4.5: Total submitted logs by entry class4.5

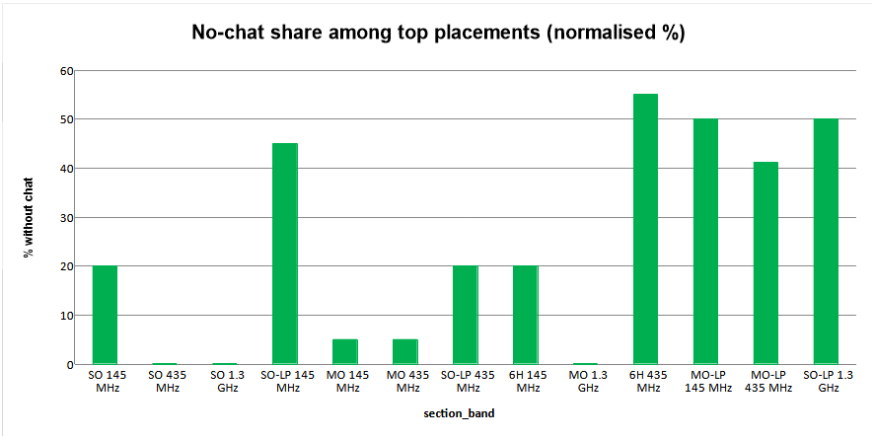


Figure 4.6: Logs without chat use placed in the top 20, by entry class4.6

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If I had to draw one conclusion from the three diagrams, it would be this: the KST chat represents the total entrants in an entry class incompletely, but among the larger classes chat use is clearly overrepresented at the top.

The reason for approaching “success through chat use” so cautiously is that, apart from our own submitted logs, I do not have the participants' log data. I cannot simply match possible sked windows for the top stations in the IARU ranking with their QSOs to identify a trend for contacts with or without chat assistance.

At DM5M the result is quite clear. Near the range limit of our station, logs show that stations were worked shortly after a sked offer, even without a positive chat reply, and that some contacts were explicitly completed via Aircraft Scatter. Across several of our logs, the analysis produced a figure of about 8% of QSOs - surprisingly high. This still does not show that those QSOs would have been impossible without the chat, even though they explicitly arose from sked requests in these cases.

Our own sample is too small for general conclusions. I therefore try to measure success factors other than IARU scores in the hope of finding a relationship by another route.

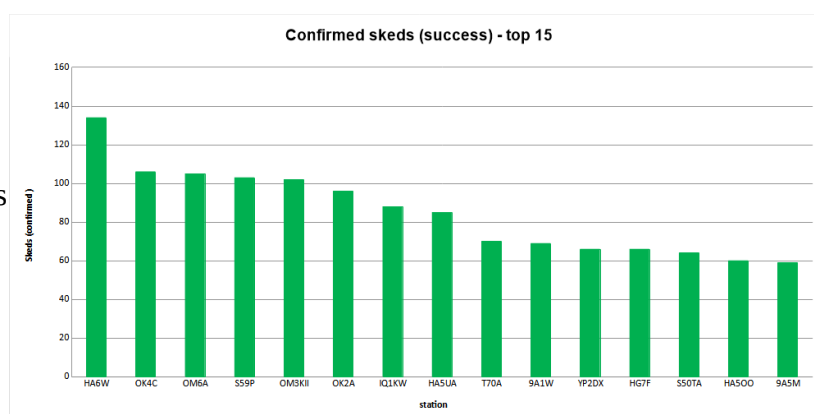
2) Relationship between contest score and “positive signals”

Section 3(c), Success and failure in the chat, describes how sked success within the chat can be defined and detected. We evaluate “positive signals” in replies to sent sked-init messages.

3)c)Success and failure in the chat

A ranking can be produced here as well. Messages with “positive signals” are conspicuously underrepresented. In comparison with the contest ranking, all callsigns are familiar, but there is no recognisable relationship with scoreboard position.

My basic assumption is that scoreboard position and chat behaviour are related. The data cannot be forced to reveal it. If the correlation is not visible, either the sample is poorly chosen or the assumption is wrong. Rather than comparing the top 15 or top 30 with the scoreboard, the next step divided the ranking into its best 20%, the next 20%, and so on.

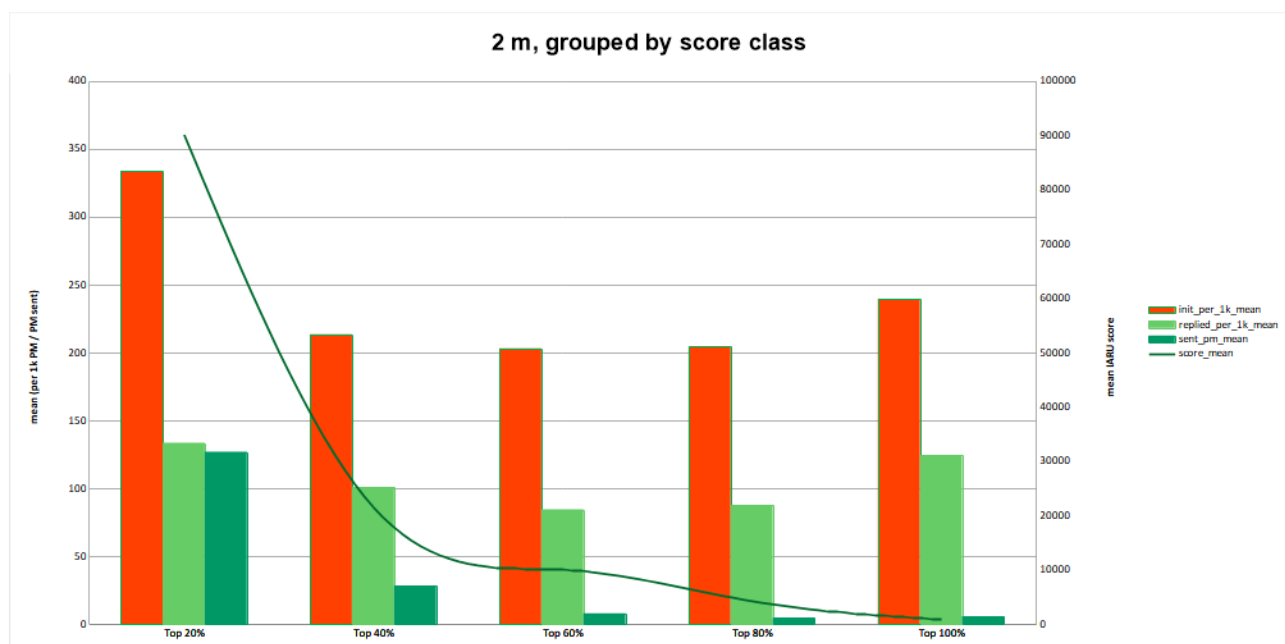


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This produces five bins, each representing 20% of an entry class's scoreboard and allowing message-sending behaviour to be compared with scoreboard position.

3) Relationship between contest score and sked initiations

It sounds unwieldy, but it is not. For the July contest, dividing results into classes produced the following picture. I show only VHF Multi Operator here. The other multi-operator classes look similar. The pattern is not always as clear in the single-operator classes, especially in the low-power classes, although it is still visible.



The stations whose behaviour correlates most strongly with scoreboard position are those that initiate many skeds and produce many sked windows from them - in other words, they elicit replies with their sked requests. The relative or peak-per-hour number of private messages is another related indicator: “As you call into the forest, so it echoes back.” Producing sked windows, and therefore a tendency to receive replies, looks like leverage.

I will offer another conclusion: the quality of the requests sent by stations that create skeds may be an indicator of obtaining more QSOs.

Put differently, sending more direct messages has an effect, but the trend is clearer when usable, contact-ready sked requests produce more replies. The interval matters too: shorter is better. It is much like a market stall. Calling loudly is not a bad idea; it attracts customers. It is even better when the offer is visible and attractive. Attractive market staff are irrelevant on the bands, so the comparison does falter. Now that we have an indicator of which stations tend to create more sked

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windows, we can ask why. Text analysis comparing the top 20% with other stations showed that stations with a high reply rate do not necessarily send “complete” opening messages. They receive replies because they contact the right stations at the right time and because their content differs. That does not mean completeness harms a sked request.

4) Recipe for success

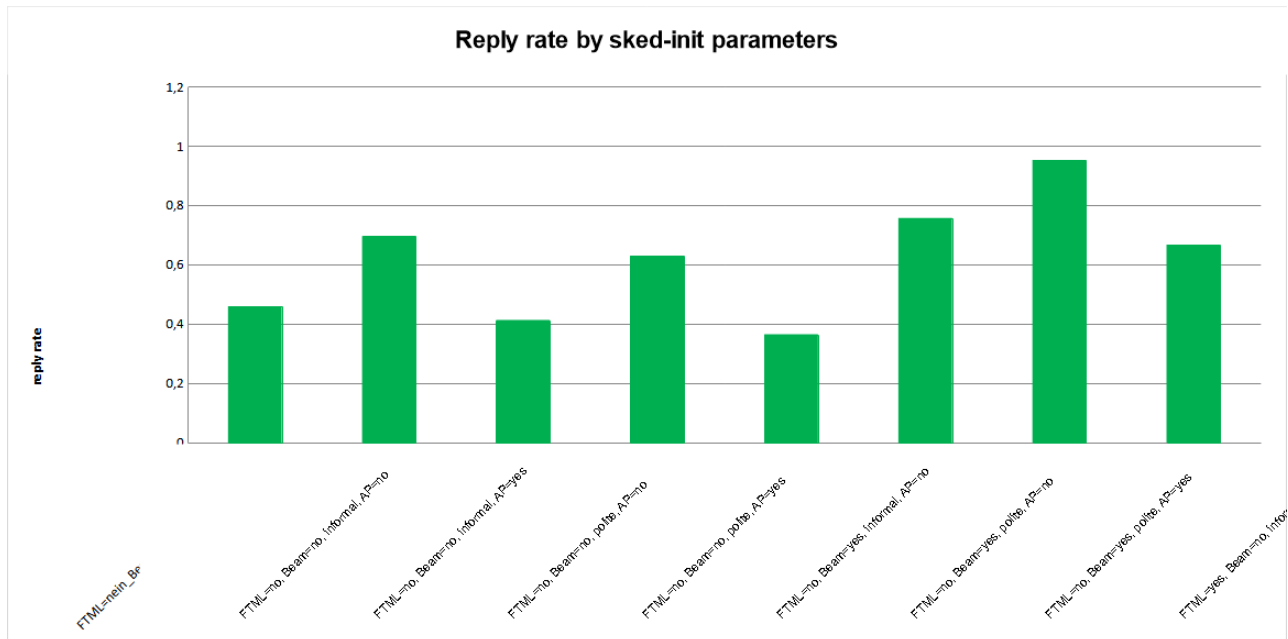
The analyses revealed a strong pattern: more immediately usable sked information leads to more replies.

On closer inspection this is not surprising. It is nevertheless useful that the correlation exists, because knowing about it can improve the workflow. As the diagram already suggests, better content makes sked requests more likely to be accepted. A well-considered request can include:

- Action or sked words such as QRG, QRV, ant, dir, call, beam, sked or AP
- A frequency in the message (“F”)
- Timing information, either a clock time or the interval until the start (“T”)
- Courtesy or anchor words such as pse, tn timer, hi (“POL”)
- A beam-direction indication (“BEAM”)
- A locator (“L”), although it does not make much difference
- An Aircraft Scatter indication (“AP”)
- The mode (“M”), which also makes little difference

Putting these factors into a diagram shows which combinations produce the most replies.

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Polite or not, AP information and an antenna-direction indication are associated with measurably better results, even if the FTML details are incomplete, need to follow later or were sent just before.

The information does not have to be present in the first request for a sked initiation to be opened. It is enough for the details to arrive close together, as they often do. When the FTML details are grouped less finely, the trend becomes particularly clear. Sked initiations with AP and beam data have an almost 30% higher reply rate than messages such as “Hi noname, try 2m?”.

They also show that preparation time has already gone into the sked. Polite or personalised requests tend to receive more replies than others.

Increasing the total number of private messages almost automatically increases the number of replies, but the relationship is not strong. This has repeatedly been visible in recent contests with S50C and OE1W, which regularly exchange first place for message volume.

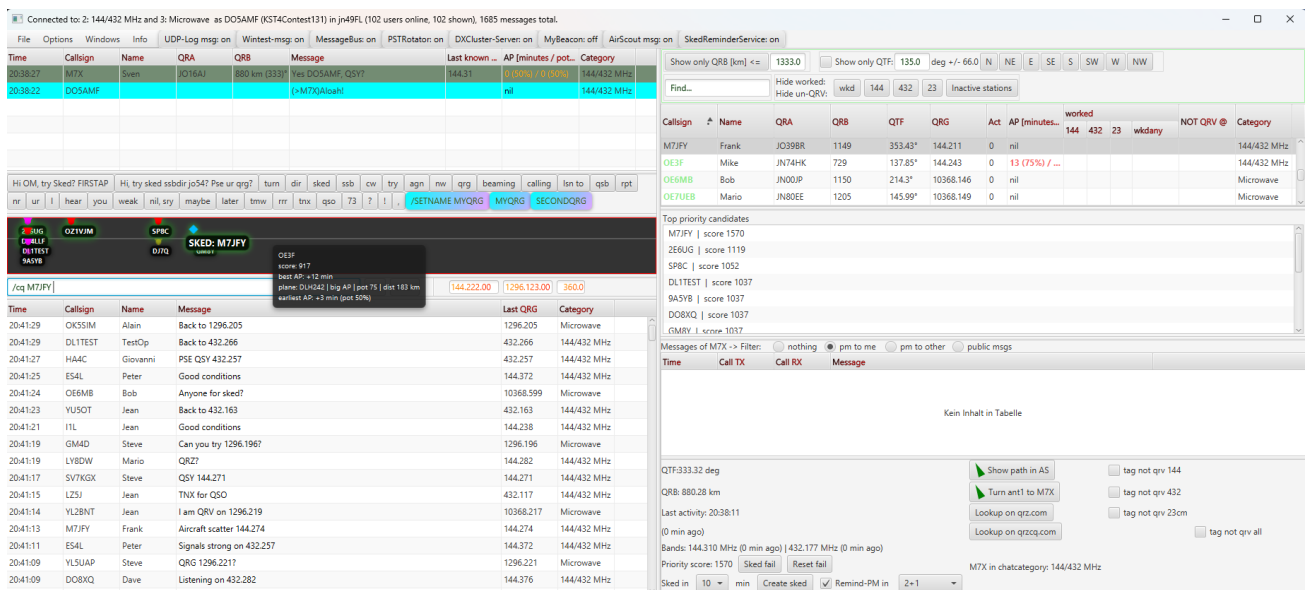
In summary, I would say that spamming does not pay. I agree with the impression described to me at the previous GHz-Tagung - I believe by Uwe, DL1SUZ - that automated messages cause more irritation than benefit. Such systems lack a heuristic that excludes stations already contacted and returning “negative signals” from the next mailing. As far as I know, VUSC cannot do that, although it does support mass mailings. This is precisely what I wanted to avoid in KST4Contest. There are now explicit ways to prevent spam.

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5) KST4Contest v1.31 update

Once the theoretical foundation had fortunately produced useful results, I could begin planning and implementing specific changes. At times the implementation overtook the theory. The latter is stubborn and time-consuming.

I will proceed as before: show a screenshot, then explain what changed, why it changed and how it works.



The 2025 GHz-Tagung material explains the structure of the user interface and its workflow-oriented design. Anyone unfamiliar with it may wish to read that paper first.

For readers who already know the interface or the program, the changes are obvious. The arrangement of some UI components may still change again.

The direction for further development was clear: sked requests had to improve and follow the statistically successful requests more closely. The goal was not to make people type messages by following a trivial checklist.

The main objective was to derive information from chat text in real time that would otherwise take a chat operator an enormous amount of effort. The information must serve the operator without creating noticeable extra work. What to do? Thus spoke Zeus.

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a) Developing a KST simulator

Over an evening beer in Dorsten I had an interesting discussion with DL8AAU, the author of wtKST, and DL2ALF, the author of AirScout, about what software could do. I always develop ideas quickly in such conversations; implementing them tends to take rather longer than I would like.

DL8AAU assumed that I had a KST server simulator. The functions had to be tested in a realistic environment, and one could hardly test everything live against the ON4KST server.

Well. Until mid-2025, that was still exactly what I did, although individual methods and functions were and are tested automatically elsewhere. I did not mention this that evening and will deny it immediately after writing these lines.

The developments that followed made a proper test server indispensable. The chat client needed a simulated environment containing plausible data. Over 31 iterations I developed a test server that produces data rubbish of limited credibility but realistic construction. It covers frequency information and its stability, reply intervals, stress tests, user mistakes and more.

b) Win-Test support

At the repeated request of several users, I addressed Win-Test support. Early in the previous year I contacted its developers and asked whether they could add the UDP ContactInfo interface used by roughly 90% of other loggers. They agreed. A few months later I asked about progress and was told that the feature was not a high priority. I could live with that.

I waited several more months, in vain. Over the course of a year I encouraged KST4Contest users to request the feature from the developers. Replies were absent or negative. Two enquiries of my own received no reply at all. I later learned that such experiences with Win-Test support are apparently not new.

Because users still wanted Win-Test support, I followed the same path I had used earlier with UCXLog, whose developer is much more open to questions. I installed Win-Test and analysed its network protocol. The result was that Win-Test's internal communication is technologically very similar to the ContactInfo interface, so I could use the packets directly. An implementation in Win-Test itself would probably have been faster still.

I am not particularly fond of the software and do not understand why so many OM's use it. I therefore postponed its integration for a long time. After the support experience, I am unlikely to become a Win-Test fan.

My final email explaining that the issue was resolved, and why, also went unanswered.

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c) A second chat category

This change follows directly from the analysis of participant presence in the chat.

The first screenshot probably made it apparent that every message table now has a new Category field. It is shown here once more.

The user list gained the same field. For the reasons described above, I added support for a second chat category in March 2025. I did not want to abandon the direction

already taken: the chat operator should be able to abstract away from categories. In my view, watching two chat windows or following two message histories is expressly undesirable. In my head, the data belongs together.

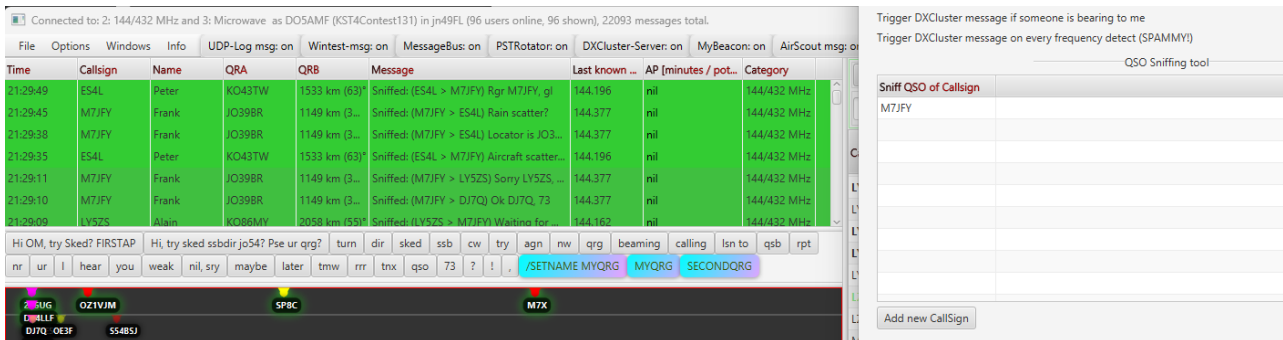
The user list can therefore contain duplicates if another station logs into both categories with the same callsign. The category is then the only difference. When I receive a message, click it and reply, the client automatically sends the reply to precisely the right sender. Messages are routed automatically. If the intended category cannot be determined, the message is sent in the category in which the recipient most recently transmitted.

Callsign	Name	QRA	QRB	QTF	QRG	Act	AP (minutes)	worked	NOT QRV	Category
LY5ZS	Alain	KO86MY	2058	55.1°	10368.181	0	nil	144 432 23		Microwave
LY7WS	Steve	KO05EB	1018	47.95°	432.599	0	nil			Microwave
LY8DW	Mario	KO35ZY	1502	53.59°	432.258	0	nil			144/432 MHz
LZ1D	Giovanni	KN29OB	1213	85.8°		1	nil			Microwave
LZ3KNE	Hans	KN23TS	1452	109.39°	10368.189	0	nil			Microwave
M31T	Jo	JO16FU	911	336.37°	1296.294	2	22 (50%)			Microwave
M7X	Sven	JO16AJ	880	333.32°	144.272	0	3 (75%) / 1...			144/432 MHz
OE3F	Mike	JN74HK	729	137.85°	144.599	0	13 (75%) / ...			144/432 MHz
OE6MB	Bob	JN00JP	1150	214.3°	10368.204	0	nil			Microwave
OE7UEB	Mario	JN80EE	1205	145.99°	1296.111	0	nil			Microwave
OE7UEB	Mario	JN80EE	1205	145.99°	432.17	0	nil			144/432 MHz
OE8Y	Hans	JN45HC	486	178.46°	10368.138	0	0 (100%) / ...			Microwave
OE8YK	Peter	JN34VW	507	185.94°	144.599	0	0 (100%) / ...			144/432 MHz
OK0J	Peter	JO35ZO	681	358.24°	144.599	0	nil			144/432 MHz

d) QSO Sniffer

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The roadmap shown at the previous GHz-Tagung already mentioned intercepting chats involving other stations. Among others, 9A2HM asked me to revisit it. He was not the first, so I implemented the idea. I had neither wanted nor imagined such a feature myself, and therefore built it as suggested by operators including SM6VTZ and as shown in the screenshot: the other stations'



The screenshot shows the KST4Contest application window. The main pane displays a table of chat messages with columns for Time, Callsign, Name, QRA, QRB, Message, Last known, AP [minutes / pot...], and Category. The messages are filtered for the 144.432 MHz frequency. The sidebar on the right shows a list of call signs, including M7JFY, and a section for 'Sniff QSO of Callsign'.

Time	Callsign	Name	QRA	QRB	Message	Last known	AP [minutes / pot...]	Category
21:29:49	ES4L	Peter	KO43TW	1533 km (63)	Sniffed: (ES4L > M7JFY) Rgr M7JFY, gl	144.196	nil	144.432 MHz
21:29:45	M7JFY	Frank	JO39BR	1149 km (3...	Sniffed: (M7JFY > ES4L) Rain scatter?	144.377	nil	144.432 MHz
21:29:38	M7JFY	Frank	JO39BR	1149 km (3...	Sniffed: (M7JFY > ES4L) Locator is JO3...	144.377	nil	144.432 MHz
21:29:35	ES4L	Peter	KO43TW	1533 km (63)	Sniffed: (ES4L > M7JFY) Aircraft scatter...	144.196	nil	144.432 MHz
21:29:11	M7JFY	Frank	JO39BR	1149 km (3...	Sniffed: (M7JFY > LY5ZS) Sorry LY5ZS, ...	144.377	nil	144.432 MHz
21:29:10	M7JFY	Frank	JO39BR	1149 km (3...	Sniffed: (M7JFY > DJ7Q) Ok DJ7Q, 73	144.377	nil	144.432 MHz
21:29:09	LY5ZS	Alain	KO86MY	2058 km (55)	Sniffed: (LY5ZS > M7JFY) Waiting for...	144.162	nil	144.432 MHz

messages appear in my inbox. The implementation took an adventurous 15 minutes.

e) Smart frequency recognition

The first objective, and the main reason for developing KST4Contest, was to build a band map from chat data. Recognising frequency strings with regular expressions cost me a fair number of nerves and had remained unchanged since version 1.

Something that had annoyed me from the beginning was the uncertainty over the band to which a frequency belonged. Operator messages such as “pse try .210” drive a developer mad. In a multi-band contest, .210 could be on any band.

That was not all. Only the last frequency mentioned for a participant was stored. The only history retained for QRG strings came from the built-in DX Cluster server, where “Sked directed to me” at least preserved a frequency for a while.

A new workflow idea arose. Would it not be useful to know, while operating, whether a selected station or the station currently being worked is QRV on another supported band as well?

Suppose I arrange a sked with such a station. I can see that other bands are possible and follow the first contact with them. That promises considerable point potential. For the misanthropes among radio amateurs, the best part is that one need not ask whether the other band is available; one can suggest the counterpart's last known QRG immediately. It was time to update the old mechanism.

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A process diagram follows in Figure 5.1, smart frequency extraction. In simplified terms, I no longer store one frequency string per station in its object. The station object now contains a Map<QRG>, allowing any number of detected frequencies to be retained. Those frequencies identify the supported bands and make a warning mechanism straightforward. When a station is logged, for example, the chat operator can immediately be warned that another band may be possible.

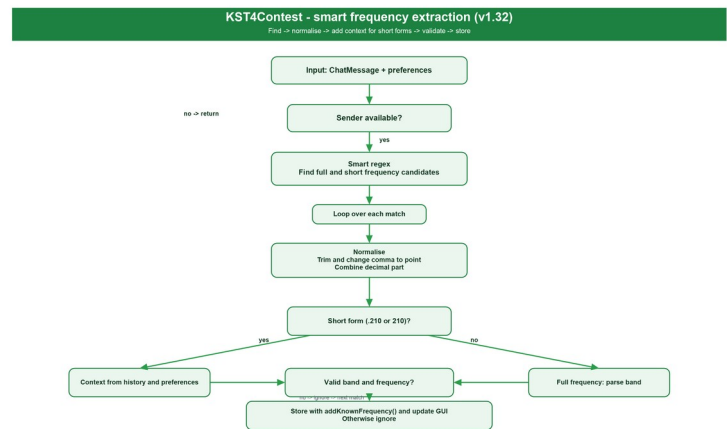


Figure 5.1: Smart frequency extraction5.1

What makes the frequency extraction smart is that, when a value such as “.210” arrives, the map is searched for a more complete frequency found during the previous 30 minutes, such as 144.210, with the same suffix and a usable prefix. If one is found, 144 can be added. The result is much more reliable than before and is usually complete without guessing the prefix. A useful side effect is that KST4Contest can generate more internal cluster spots, because that happens - and only makes sense - when the QRG has been recognised completely.

Callsign prefixes and suffixes such as -2 or -70 are abstracted away as well. Frequencies are stored for the normalised callsign and retained in every callsign object that shares the raw base string.

It works well in the simulation; practice still has to prove it. If that does not enable a more deliberate sked request, what would?

f) Priority Score and its consequences

Now, at last, this becomes genuinely innovative again - at least I think so.

Most existing chat clients with chat-room operation share the same basic design and centre on the user list within a channel. The well-seasoned radio amateur knows the idea from programs such as mIRC in the good old IRC days, or AOL chat rooms.

There is nothing inherently wrong with the principle, but it is not especially practical. In the days of IRC and AOL chat rooms, we trained ourselves to type quickly in our youth, and it made almost no difference whose ear we talked off.

The user still sits in front of a list and decides whom to contact next. In the KST chat, the same applies. The operator uses a mental heuristic to decide which station might be the most sensible

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next sked partner. The broadcast chat sometimes helps, as can favourable aircraft crossings, although these have to be looked up in the user list. I noticed that this decision does not become easier as a contest grows longer. I therefore decided that stations in the user list needed an objective measure of suitability. I call it the “Priority Score”, based on a station metric. The user list becomes a data source for the program rather than the centre of every manual decision. It still has to prove itself, but my present idea is that one should search the user list only when one is very, very bored. Figure 5.2: Score process. Figure 5.3: Score factors.

Calculating the score is a larger process with several inputs and triggers. Station metrics are created for a clean implementation and collected in a ScoreSnapshot. When a relevant change triggers recomputation, the result is updated in the UI. The process diagram appears above and the score factors below. The window shows the resulting priority list, which I rather boldly hope will replace the user list. I created a manual sked for M7JFY beforehand, which is why the station appears so high. ScoreService recognises time-critical situations and brings those stations to the top. It also considers activity, reply behaviour, calling direction, “positive signals” from text analysis, per-band Worked status, Aircraft Scatter status, beam direction and distance. A medium distance scores more highly than a very high or low distance because it usually offers the best ratio of points to effort and can be worked quickly. Failed skeds reduce the score, as do unanswered messages. This actively avoids prioritising stations when the path is already known not to work or the station is unlikely to reply.

Score / priority process (KST4Contest)

Brief, without detail - keywords for the explanation

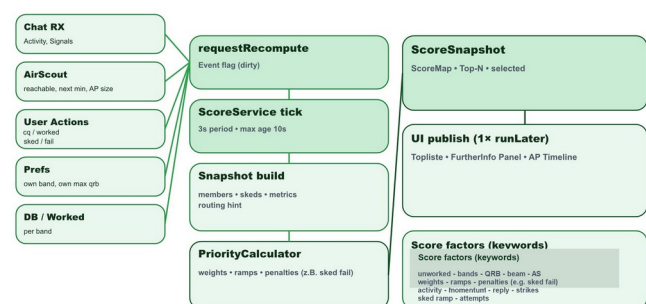


Figure 5.2: Score process

Score-Bewertungsfaktoren (KST4Contest)

Stichworte • Kurzbeschreibung • basiert auf der aktuellen PriorityScore-Look

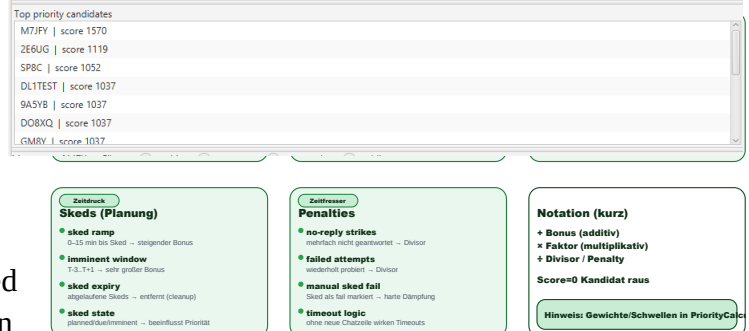


Figure 5.3: Score factors

g) Skeds

Figure 5.4: Sked process. One feature I had always missed was a reminder for skeds. From JO51IJ, SP2SOO can be

Sked lifecycle: time window -> actions -> score effect

Skeds shape what comes next in the chat: ramp, imminent, boost and expiry.

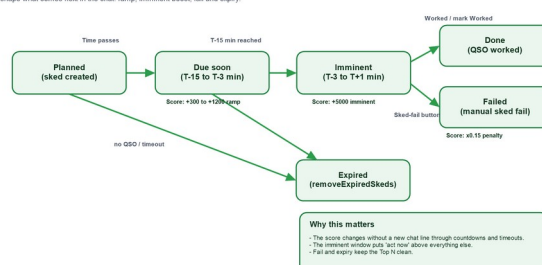


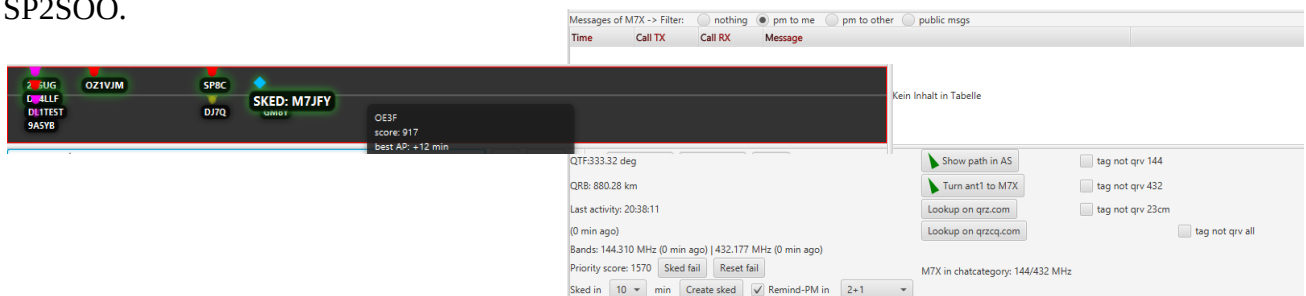
Figure 5.4: Sked process

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reached by aircraft only about every 30 minutes, and useful opportunities are rare. When the next aircraft is still some way off, one easily promises the other operator - and oneself - to return when it arrives. In the heat of the chat, the moment is missed after all. That is discourteous and also annoying when the opportunity simply passes. What could be more obvious than a sked timer, preferably one that can remind not only me but also the station to be worked, with messages sent in good time?

The window on the right contains such a function. The sked time can be selected within limits. Clicking “Create sked” arms the short-term alarm. “Remind PM in” is enabled when the other operator should receive a private message n minutes before the sked - in this case two minutes and again one minute before the scheduled time.

After “Create sked”, the station receives very high priority boosts at defined intervals, depending on how far in the future the sked lies, and therefore leads the score list. Visual and audible warnings announce a manually arranged sked. With luck, I will not miss another sked, and neither will SP2SOO.



This is how a sked appears in the timeline, with about seven minutes remaining.

h) Aircraft Scatter timeline

While enjoying the beer mentioned above after the previous GHz-Tagung, DL8AAU expressed a heartfelt wish: it would be useful to “pick out” counterpart stations according to the aircraft. I immediately had ideas, several of which are being implemented. The subject can become quite complex, depending on how it is approached.

My original idea was to build a matrix of stations and store sets of counterpart stations containing Aircraft Scatter data as station properties. Because of complexity and legal considerations, flight-path calculation would be decentralised as cluster computing across the distributed KST4Contest clients. A service at hamradioonline.de would process and cache the data and make it available to other chat clients. It works, but it is not active.

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One problem is the often venerable hardware used by radio amateurs. The cluster-computing master distributes Aircraft Scatter calculations to clients and uses their results. The number of paths a client can calculate varies enormously, and the original implementation underestimated that variation. My development machine at home is naturally fairly powerful; others are not. Calculations take too long, and synchronisation problems arise between AirScout and the chat client because results arrive much later than expected. There appears to be a computation-request queue. I have designed a solution but not implemented it. It concerns the timing of work distribution. AirScout was not built for such contortions. The overall processing load is not the problem; the delay of individual jobs is.

To cut a long story short, there is a simpler way. I call the quick local solution, which does not need a large user community, the Aircraft Scatter timeline. It is a time axis showing high-Priority-Score stations in the antenna direction, based on integrated AirScout data.

For each minute that can be displayed, high-priority stations appear when an Aircraft Scatter window exists. Because there are very often many aircraft for one target, the timeline prefers aircraft with high reflection potential. Only when no high potentials exist are other aircraft shown with their highest-priority stations, and never more than four simultaneously per minute. Figure 5.5: Aircraft Scatter timeline process.

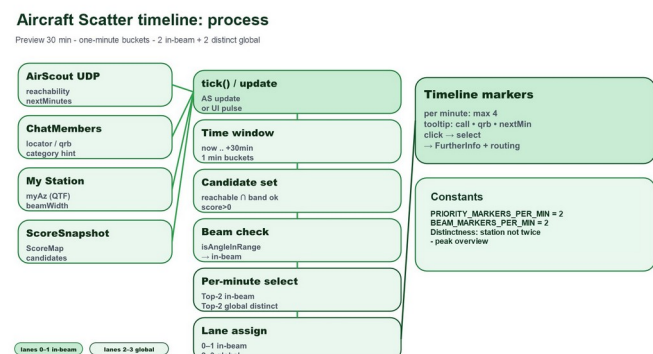
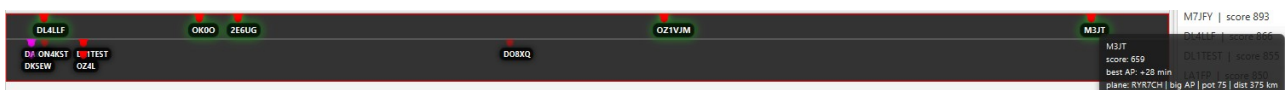


Figure 5.5: Aircraft Scatter timeline process

The timeline also uses the PSTRotator interface. If the rotor points at a station within the configured beamwidth, that station lights up strongly in the timeline. The following screenshot shows an example; DO8XQ is the only station outside the beam.



If the rotor does not point in that direction, the station appears transparent, as DO8XQ does.

The marker colours, called “crystals”, represent the aircraft's reflection potential. The four lanes reflect the current rotor position, as shown in the diagram.

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Because the Priority Score already considers whether a station recently requested a sked in my direction, and because its frequencies are usually known as another input to the score, it should be possible to click through these stations and work them one after another. March will show how well that works.

Here too, the initial analysis has direct consequences. The heuristic makes it possible to send high-quality sked requests that already contain the information needed to start immediately.

i) PSTRotator interface

Last year I mentioned that I had integrated the interface despite not owning such a rotor. I returned to the subject unexpectedly after my good friend Jörg, DK8DE, and I spent considerable time upgrading and modifying rotors, alongside other projects such as the sequencer with CW buffer function that we christened “pro buffer”. Both provide material for another presentation and will not be pursued here. Jörg will present the sequencer in Dorsten in 2026.

The rotor interface will be active from the next release. It has been tested with my old Spid RAS. We also found and handled some uncertainties that Spid supplies free of charge with its protocol. The visible result is an innocent field to the right of the text input; the side effects reach the Aircraft Scatter timeline and the user-priority table. The displayed 360° is the rotor's initial value. To its left is the category 2 QRG, and to the left of that the category 1 QRG. Automatically enabling user-list filters from the rotor position would also be conceivable, but it is not currently implemented and



The screenshot shows a horizontal interface element. On the left is a text input field. To its right are two buttons labeled 'TX' and 'clear'. Further right are three rectangular boxes displaying frequency values: '144.222.00', '1296.123.00', and '360.0'.

may belong on the roadmap.

j) Auto-answer

Last year I mentioned that logging a station could send an automatic 73 message containing a checksum for cross-checking. A murmur filled the room. It was some time ago, but I believe the consensus was that this would not comply with the contest rules. I therefore decided not to implement such a feature, just as I avoided mass mailings. In light of the analysis above, a 73/tnx packet as confirmation would nevertheless be a neat idea.

I did not implement that either. What I did implement is an automatic reply to QRG questions and a general reply for operators who do not want to be disturbed. KST4Contest can automatically answer common frequency-request phrases with both QRGs.

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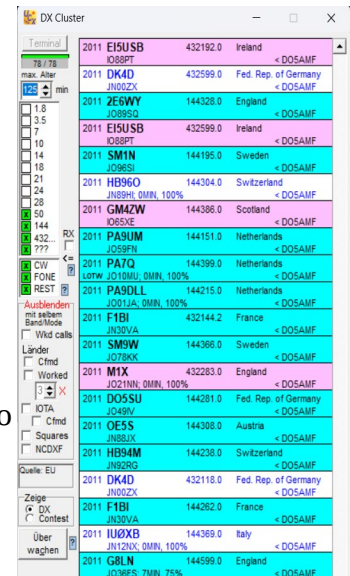
I) DX Cluster server and AP information integration

The DX Cluster server informs the contest logger like a normal cluster. The feature was already present last year. Feedback has been very positive, and the intended sequence - click, work, click the next one - actually works for about 50% of spots, depending on conditions.

The spots are generated by the “Sked directed to me” mechanism.

In the previous release I added information about the next aircraft passage to the locator in the spot, giving the recipient as much usable information as possible.

The screenshot shows DX Cluster messages in UCXLog. One click also changes the TRX QRG, making it convenient to step through potential counterpart stations.



Call	Frequency	Country	Mode
2011 EISUSB	432192.0	Ireland	< DOSAMF
2011 DK4D	432599.0	Fed. Rep. of Germany	< DOSAMF
2011 26WVY	144328.0	England	< DOSAMF
2011 EISUSB	432599.0	Ireland	< DOSAMF
2011 SM1N	144195.0	Sweden	< DOSAMF
2011 HB96O	144304.0	Switzerland	< DOSAMF
2011 GM4ZW	144386.0	Scotland	< DOSAMF
2011 PA9UM	144151.0	Netherlands	< DOSAMF
2011 PA7Q	144399.0	Netherlands	< DOSAMF
2011 PASDLL	144215.0	Netherlands	< DOSAMF
2011 F1BI	432144.2	France	< DOSAMF
2011 SM9W	144366.0	Sweden	< DOSAMF
2011 M1X	432283.0	England	< DOSAMF
2011 DO5SU	144281.0	Fed. Rep. of Germany	< DOSAMF
2011 OE5S	144388.0	Austria	< DOSAMF
2011 HB94M	144238.0	Switzerland	< DOSAMF
2011 DK4D	432118.0	Fed. Rep. of Germany	< DOSAMF
2011 F1BI	144262.0	France	< DOSAMF
2011 IU0XB	144369.0	Italy	< DOSAMF
2011 GR1N	144599.0	England	< DOSAMF

6) Integrating the software into contest operation

I will quote myself from the previous year:

“As I have since put it, contesting today differs greatly from the way I experienced it at the beginning of my contesting life. I am a computer scientist and therefore have an intrinsic urge to play with technology. At the same time, I understand the views of testers who were and are successful without a chat and do not want their radio operation to depend on a medium other than radio itself.

The chat is something one has to embrace. That is a process inherent in our hobby. For the testers among us, a 24-hour event is the ideal time to test our own resilience and that of the equipment.”

That was my conclusion then, and it remains my conclusion this year. Even if my statistical reasoning is not presented as impeccably as my much-appreciated Professor Dr Frank Nürnberg would surely prefer, I consider it demonstrable that using the chat has a noticeable effect on contest results. Even if a good placing is more of a side effect, there is certainly no harm in processing skeds as efficiently as possible.

PS: I will, of course, gladly provide the source data for the analyses in the opening sections to anyone interested.

I look forward to the March contest, when the latest functions can be tested under real operating conditions, and I am curious whether using the client will affect our score.

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7) To-do list

At this point I have copied an extract from my current working notes.

- Time-window analysis, which was already on last year's list
 - Warn in the client when a station is QRV that is usually available only briefly
 - The concept is complete, but it requires a server and will therefore take a little longer
- Logbook-history analysis
 - Warn in the client when a station is QRV that has usually already been worked
 - DK5OH suggested this, and it is one of the next objectives
- Line wrapping, also carried over from last year
 - Wrap long text and make it possible to copy
- “Auto 73”, for confirmation only
 - Would that be more acceptable?
- “Back to CQ QRG” spot after QSY
 - After QSY, create an internal self-spot for the former CQ QRG so that the TRX can return to it without being touched
 - Further spots are planned, including a spot on sked to automate the QRG
- Protocol and web service for exchanging chat users' QRG and QRV-band information
 - Work with G1YFG to develop an exchange protocol for QRG data
 - This is pending for the reasons explained above
- Aircraft and category filters for the user list
- Locator Sniffer and Keyword Sniffer
- Lifetime for logged flags
 - At present the user still has to reset them manually; that has to go
- Manual Worked button
 - For the occasion when a UDP packet goes missing...

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8) Facts and figures

a) Development time and effort

The chat client has been under development since mid-2022, usually after the children are in bed and for no more than two or three hours a day. Without exaggeration, hundreds of hours have gone into it. Creating the user manual sounds trivial, yet even the first version of the client required 22 net hours. The manual needs a comprehensive update because almost nothing has happened there for about a year.

b) Thanks

I owe the fact that the chat client ever left the development environment to Konrad Neitzel, DC9DJ. I only learned that he was a radio amateur after approaching him, as a professional software developer from the Java forum, with my deployment problem. I am very grateful. There are also users who regularly test the software before publication and report the results. Many thanks to them as well.

c) Size of the project

The complete project currently contains about 45,000 lines of code. It has barely grown since last year because I have also done a great deal of refactoring. The user community is steadily increasing and, according to the update requests, has been in three figures for some time. I regularly receive requests and ideas from users, and I am grateful for those as well.

d) It will not make you rich

The chat client is primarily developed for DM5M and for my own use, so that is not a problem. But since people say one should not talk about money, and I am a rebel, I will do so anyway. At the time of writing, the grand total of contributions was:

- **EUR 80**

Well, there we are. If you use the software, like it or want to support further development, the application and its user manual explain how. Contributions of code are welcome too.

The source code is publicly available. At the time of the presentation, the public repository trailed my local development state.

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9) Download

At the time of the presentation, the chat client was available from:

- <https://do5amf.funkerportal.de/>

The download was provided there.

I planned to complete and release version 1.31, containing the new functions described here, by the time of the presentation. Several testers were already waiting, but a number of items still remained. After registering a domain at the end of 2024 for the planned QRG and QRV-data web service, I also intended to move the downloads to:

- <https://hamradioonline.de/>

I expected both addresses to remain available for some time.

For questions about the client: praktimarc+kst4contest@gmail.com

73 de DO5AMF, Marc